

A method to determine the minimal null control time of one-dimensional linear hyperbolic systems

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How fast can a one-dimensional transport process be brought to rest by acting on just one end? This is a question about the *minimal null control time*, a central notion in control theory that, for coupled hyperbolic systems, turns out to be delicate.

Here, we consider first-order linear hyperbolic systems of the form:

$$\begin{cases} \frac{\partial y}{\partial t}(t, x) + \Lambda(x) \frac{\partial y}{\partial x}(t, x) = M(x)y(t, x), \\ y_-(t, 1) = u(t), \quad y_+(t, 0) = Qy_-(t, 0), \\ y(0, x) = y^0(x), \end{cases} \quad (t, x) \in (0, T) \times (0, 1).$$

The components propagate at the speeds prescribed by Λ , interact through the internal coupling M , are coupled at the boundary through Q , and are steered by a control u acting on one side. Such systems arise as linearizations, around steady states, of the nonlinear balance laws governing many transport phenomena [BC16; YK22]. Already in this linear setting, determining the minimal control time is a subtle problem, as it depends not only on the propagation speeds but on the couplings.

In this talk, we present a general method that computes this minimal control time exactly, whenever Λ and M are sufficiently smooth functions of x . This presentation is based on the joint work [HO25] with Long Hu[†].

References

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